

So, the required point is (0, 9).

Let us check our solution : $AP = \sqrt{(6-0)^2 + (5-9)^2} = \sqrt{36+16} = \sqrt{52}$

$$BP = \sqrt{(-4-0)^2 + (3-9)^2} = \sqrt{16+36} = \sqrt{52}$$

Note : Using the remark above, we see that (0, 9) is the intersection of the y-axis and the perpendicular bisector of AB.

EXERCISE 7.1

- Find the distance between the following pairs of points :
(i) (2, 3), (4, 1) (ii) (-5, 7), (-1, 3) (iii) (a, b), (-a, -b)
- Find the distance between the points (0, 0) and (36, 15). Can you now find the distance between the two towns A and B discussed in Section 7.2.
- Determine if the points (1, 5), (2, 3) and (-2, -11) are collinear.
- Check whether (5, -2), (6, 4) and (7, -2) are the vertices of an isosceles triangle.

- In a classroom, 4 friends are seated at the points A, B, C and D as shown in Fig. 7.8. Champa and Chameli walk into the class and after observing for a few minutes Champa asks Chameli, "Don't you think ABCD is a square?" Chameli disagrees. Using distance formula, find which of them is correct.

- Name the type of quadrilateral formed, if any, by the following points, and give reasons for your answer:

- (-1, -2), (1, 0), (-1, 2), (-3, 0)
- (-3, 5), (3, 1), (0, 3), (-1, -4)
- (4, 5), (7, 6), (4, 3), (1, 2)

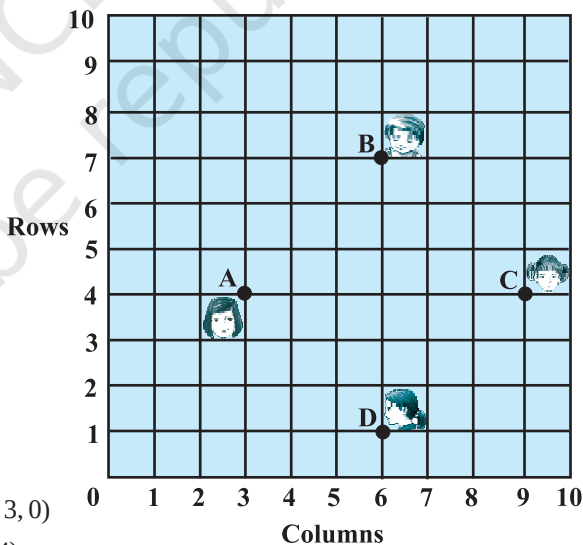


Fig. 7.8

- Find the point on the x-axis which is equidistant from (2, -5) and (-2, 9).
- Find the values of y for which the distance between the points P(2, -3) and Q(10, y) is 10 units.

EXERCISE 7.2

1. Find the coordinates of the point which divides the join of $(-1, 7)$ and $(4, -3)$ in the ratio $2 : 3$.
2. Find the coordinates of the points of trisection of the line segment joining $(4, -1)$ and $(-2, -3)$.

3. To conduct Sports Day activities, in your rectangular shaped school ground ABCD, lines have been drawn with chalk powder at a distance of 1m each. 100 flower pots have been placed at a distance of 1m from each other along AD, as shown

in Fig. 7.12. Niharika runs $\frac{1}{4}$ th the

distance AD on the 2nd line and

posts a green flag. Preet runs $\frac{1}{5}$ th

the distance AD on the eighth line

and posts a red flag. What is the

distance between both the flags? If

Rashmi has to post a blue flag exactly

halfway between the line segment

joining the two flags, where should

she post her flag?

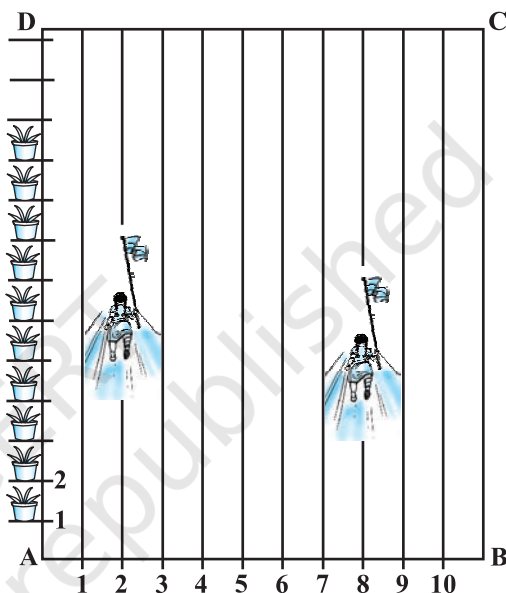


Fig. 7.12

4. Find the ratio in which the line segment joining the points $(-3, 10)$ and $(6, -8)$ is divided by $(-1, 6)$.
5. Find the ratio in which the line segment joining $A(1, -5)$ and $B(-4, 5)$ is divided by the x -axis. Also find the coordinates of the point of division.
6. If $(1, 2)$, $(4, y)$, $(x, 6)$ and $(3, 5)$ are the vertices of a parallelogram taken in order, find x and y .
7. Find the coordinates of a point A, where AB is the diameter of a circle whose centre is $(2, -3)$ and B is $(1, 4)$.
8. If A and B are $(-2, -2)$ and $(2, -4)$, respectively, find the coordinates of P such that $AP = \frac{3}{7} AB$ and P lies on the line segment AB.
9. Find the coordinates of the points which divide the line segment joining $A(-2, 2)$ and $B(2, 8)$ into four equal parts.
10. Find the area of a rhombus if its vertices are $(3, 0)$, $(4, 5)$, $(-1, 4)$ and $(-2, -1)$ taken in order. [Hint : Area of a rhombus = $\frac{1}{2}$ (product of its diagonals)]